Workshop

Anomaly Detection

1. Density-based Anomalies are instances that are in regions of low density or low local/relative density. This needs a density estimator such as Kernel Density Estimator (KDE):

$$KDE(x|D) = \frac{1}{|D|}\sum_{y \in D} \kappa(x, y)$$

2. OCSVM learns a boundary in high dimensional space of a kernel to separate between all training points (as normal points) and the origin (as anomaly) by learning a hyperplane (using an optimization algorithm).
   - The output can be interpreted as the probability being anomaly.

Where is the distance function used?

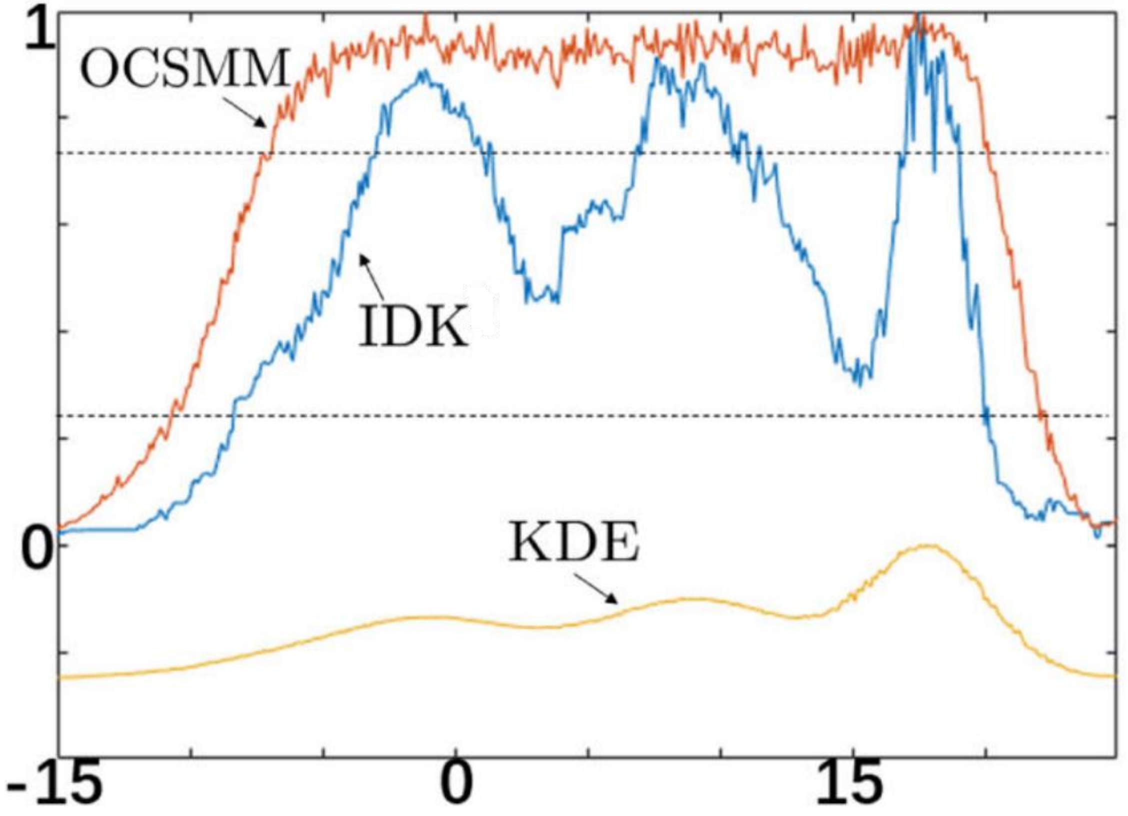
Data independent distance/similarity measures

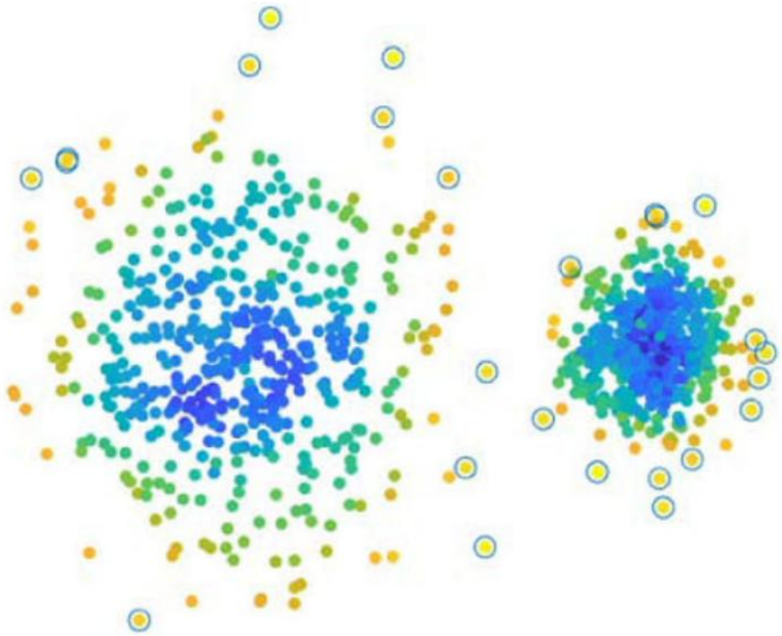
Purely based on the geometry position of the points in the data/input space, independent of data distribution.

- L_p Norm (Minkowski distance): $\boldsymbol{Dist(X,Y)} = \left(\sum_{i=1}^{d} |x_i - y_i|^p\right)^{1/p}$
- Cosine similarity: cosine($\mathbf{X_1}$, $\mathbf{X_2}$) = $<\mathbf{X_1},\mathbf{X_2}> / \|\mathbf{X}_1\|\, \|\mathbf{X}_2\|$
- Kernel functions: Gaussian kernel, Laplacian kernel, exponential kernel

$$\text{Gaussian kernel: } \kappa(x, y) = \frac{1}{(\sigma\sqrt{2\pi})^d} \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right).$$

The dataset has 3 Gaussian distributions with increasing densities
Explain how (i) KDE & (ii) OCSVM detects anomalies in this dataset





Isolation Distributional Kernel (IDK)

Definition 1. *Isolation Distributional Kernel of two distributions $\mathcal{P}_S$ and $\mathcal{P}_T$ is given as:*

$$\begin{aligned}
\widehat{\mathcal{K}}_I(\mathcal{P}_S, \mathcal{P}_T|D) &= \frac{1}{|S||T|}\sum_{x\in S}\sum_{y\in T}\kappa_I(x,y|D) \\
&= \frac{1}{t|S||T|}\sum_{x\in S}\sum_{y\in T}\langle\Phi(x|D),\Phi(y|D)\rangle \\
&= \frac{1}{t}\left\langle\widehat{\Phi}(\mathcal{P}_S|D),\widehat{\Phi}(\mathcal{P}_T|D)\right\rangle
\end{aligned}$$

where $\widehat{\Phi}(\mathcal{P}_S|D) = \frac{1}{|S|}\sum_{x\in S}\Phi(x|D)$ is the empirical feature map of the kernel mean embedding; and Isolation Kernel: $\kappa_I(x,y|D) = \frac{1}{t}\langle\Phi(x|D),\Phi(y|D)\rangle$.

IDK method for point anomaly detection

The idea is to use IDK to measure the similarity effectively between every point (in the given dataset D) and D (the reference dataset).

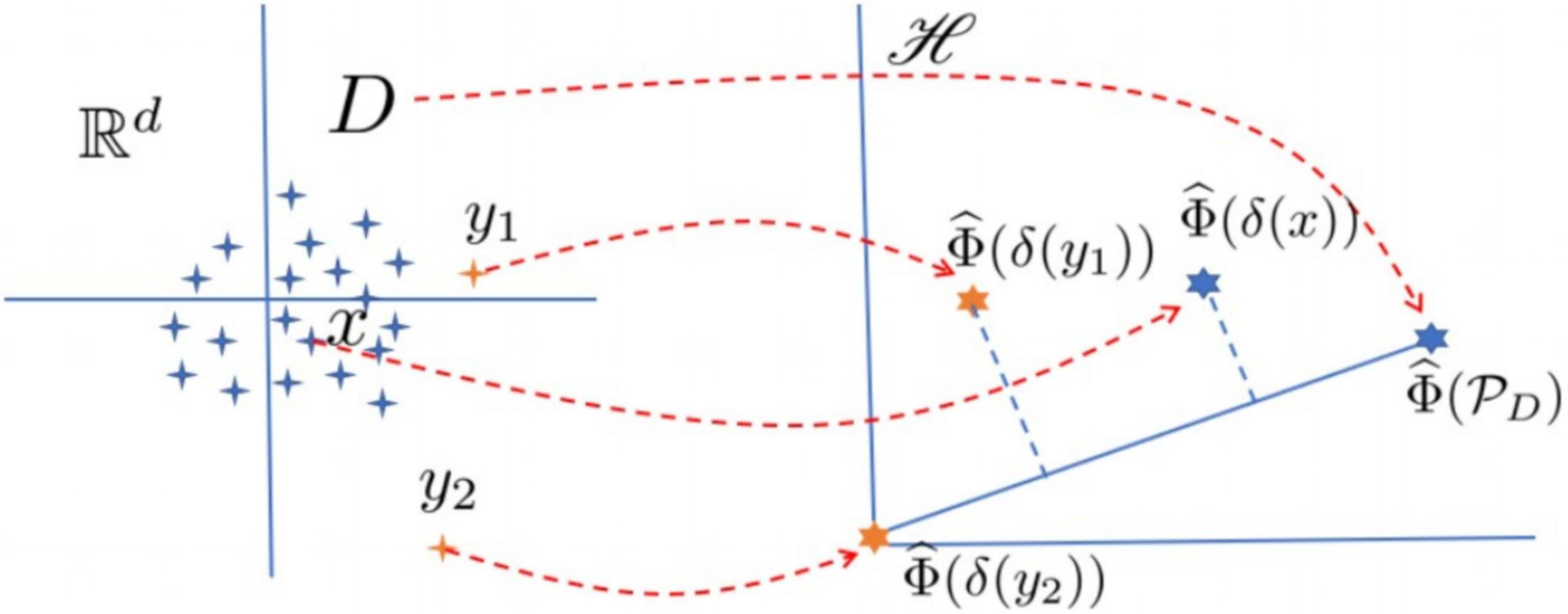
- If $x \sim \mathcal{P}_D$, $\widehat{\mathcal{K}}(\delta(x), \mathcal{P}_D)$ is large, which can be interpreted as x is likely to be part of $\mathcal{P}_D$.
- If $y \nsim \mathcal{P}_D$, $\widehat{\mathcal{K}}(\delta(y), \mathcal{P}_D)$ is small, which can be interpreted as y is not likely to be part of $\mathcal{P}_D$.

Definition of anomaly：

> 'Given a similarity measure $\widehat{\mathcal{K}}$ of two distributions, an anomaly is an observation whose Dirac measure δ has a low similarity with the distribution from which a reference dataset is generated.'

Example computation

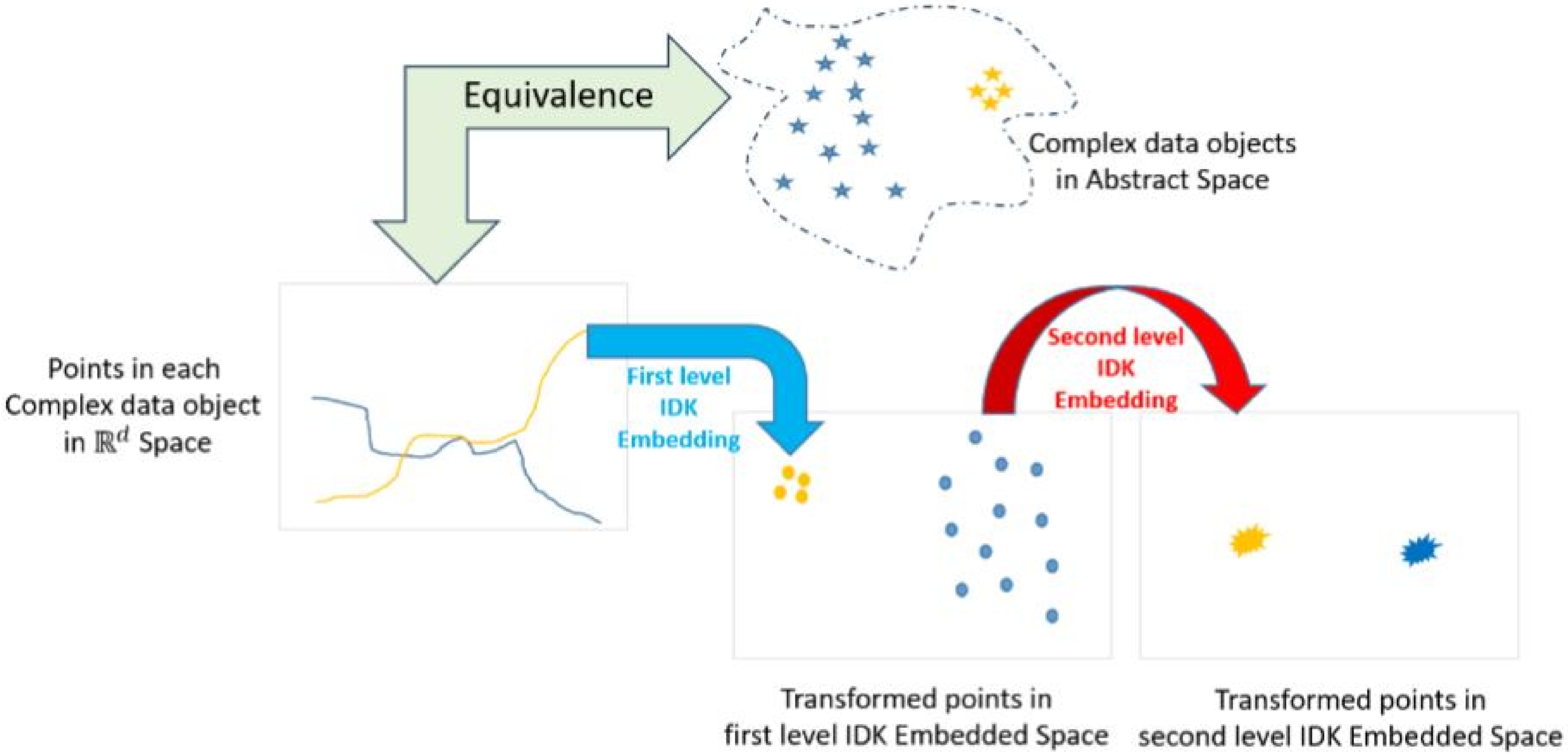
$$\widehat{\mathcal{K}}_I(\delta(x), \mathcal{P}_D) \quad = \quad \frac{1}{t}\left\langle \widehat{\Phi}(\delta(x)), \widehat{\Phi}(\mathcal{P}_D) \right\rangle$$



Discussion

1. Explain how IDK detects anomalies in this dataset
2. What if the dataset has three Gaussian distributions of the same variance?
3. Explain the key differences and similarities between IDK and KDE/OCSVM
4. Summarize what you have learned

Using two levels of distributional information to solve problems involving complex data objects



Two Insights

Insight 5. *Points in $\mathbb{R}^d$, of each complex data object in the abstract space, have some dependency that violates the i.i.d. assumption in $\mathbb{R}^d$. Yet, the points can be treated as those generated by some restricted form of distribution that complies with the dependency. This treatment enables a distributional kernel to be used for embedding and the similarity measurement between two complex data objects.*

Insight 6. *A group of complex data objects in an abstract space can be viewed as i.i.d. sampled objects generated from an unknown distribution.*

What is Group Anomaly Detection?

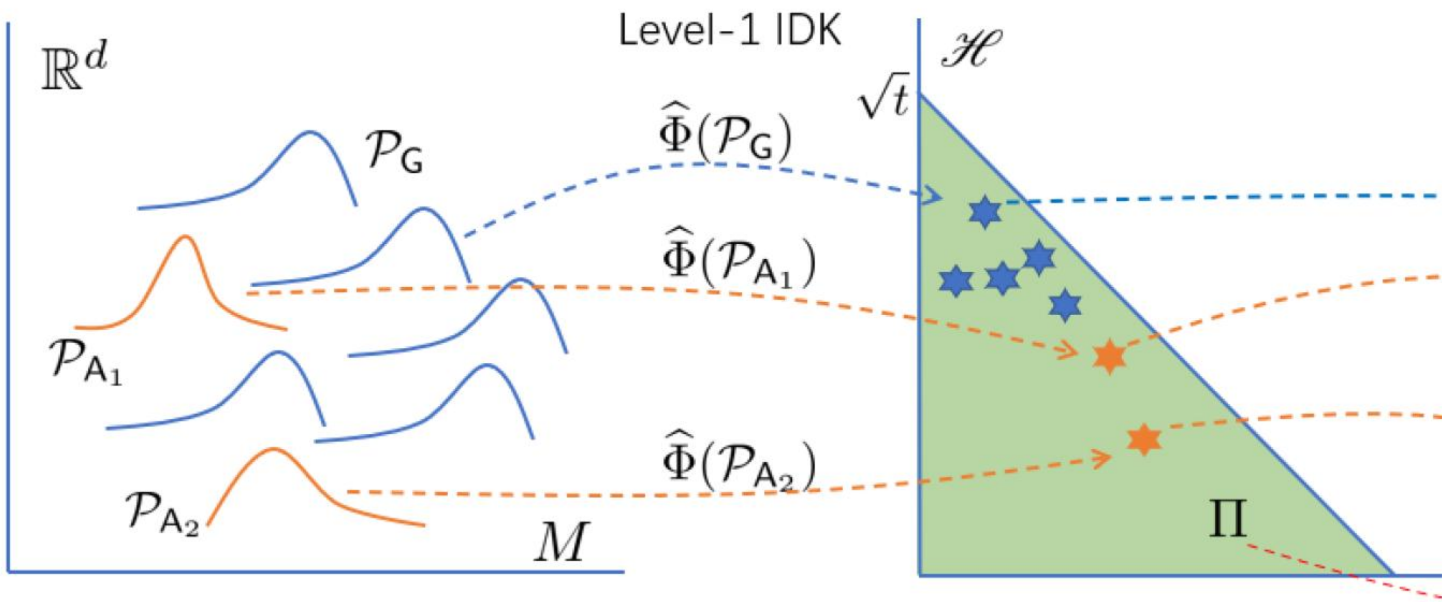
- Group anomaly detection aims to detect group anomalies which exist in a dataset of groups of data points, where no labels are available, in an unsupervised learning context.
- Group anomaly detection requires a means to determine whether two groups of data points are generated from the same distribution or from different distributions.

Group Anomaly Detection using IDK^2 [#]: Geometrical interpretation of IDK mappings

IDK^2 : Using two levels of IDK to detect group anomalies

Level–1 IDK maps each group to a point in Level–1 Hilbert space

Level–2 IDK performs point anomaly detection in Level–2 Hilbert space



[#] K.M. Ting, B.-C. Xu, T. Washio, Z.-H. Zhou. Isolation Distributional Kernel: A New Tool for Point and Group Anomaly Detection. IEEE Transactions on Knowledge and Data Engineering (2023).

Commonly Used Trajectory Distances

- Hausdorff distance
- DTW distance
- Frechet distance

- How can you use OCSVM or NN detector or LOF to detect anomalous trajectories in a dataset of trajectories?

Hausdorff distance (from Wikipedia)

Definition [edit]

Let (M, d) be a metric space. For each pair of non-empty subsets $X \subset M$ and $Y \subset M$, the Hausdorff distance between X and Y is defined as

$$d_{\mathrm{H}}(X,Y) := \max\left\{\sup_{x\in X} d(x,Y),\ \sup_{y\in Y} d(X,y)\right\},$$

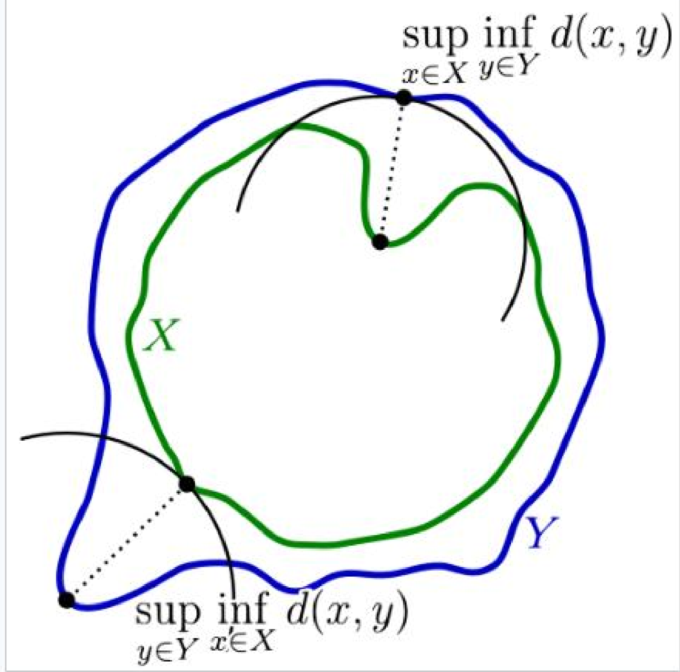
where $\sup$ represents the supremum operator, $\inf$ the infimum operator, and where $d(a,B) := \inf_{b\in B} d(a,b)$ quantifies the distance from a point $a \in X$ to the subset $B \subseteq X$.

An equivalent definition is as follows.[3] For each set $X \subset M$, let

$$X_\varepsilon := \bigcup_{x\in X}\{z\in M \mid d(z,x)\le\varepsilon\},$$

which is the set of all points within ε of the set X (sometimes called the ε-fattening of X or a generalized ball of radius ε around X). Then, the Hausdorff distance between X and Y is defined as

$$d_H(X,Y) := \inf\{\varepsilon \ge 0 \mid X \subseteq Y_\varepsilon \text{ and } Y \subseteq X_\varepsilon\}.$$



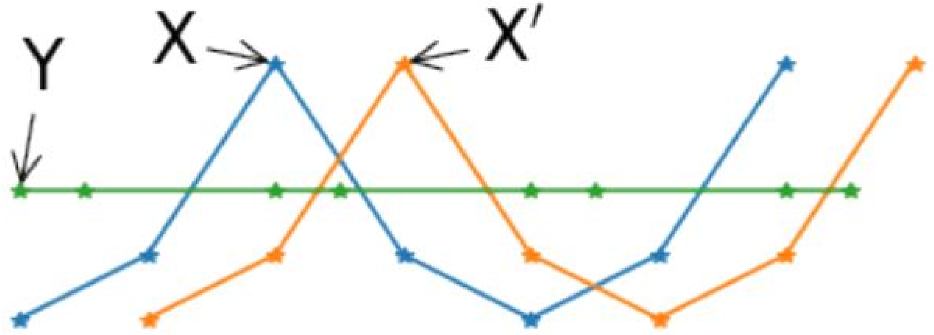
Components of the calculation of the Hausdorff distance between the green curve X and the blue curve Y.

Properties of different measures

	$\mathcal{K}_I$	$\mathcal{K}_G$	d_W	d_H	d_F
Uniqueness property	✓	✓	$\times$	$\times$	$\times$
Point-to-point distance-based	$\times$	✓	✓	✓	✓
Distribution-based	✓	✓	$\times$	$\times$	$\times$
Data-dependent	✓	$\times$	$\times$	$\times$	$\times$
Time complexity	$m+n$		mn	mn	$mn\log(mn)$

Uniqueness: Point-to-point based Distance vs Distribution-based kernel

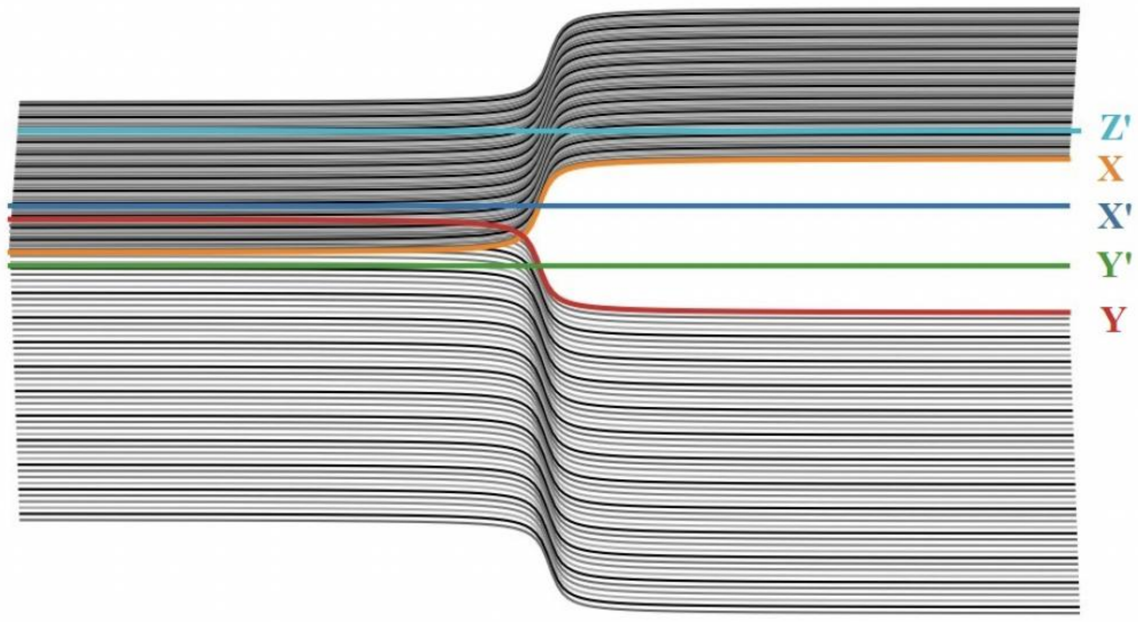
- Uniqueness property, i.e., $dist(X,Y) = 0$ if and only if $X = Y$.
- d_W, d_H, d_F: based on point-to-point matching without considering the distribution of points. Do not have the uniqueness property.
- $\mathcal{K}_I, \mathcal{K}_G$: based on kernel mean embedding. Have the uniqueness property.

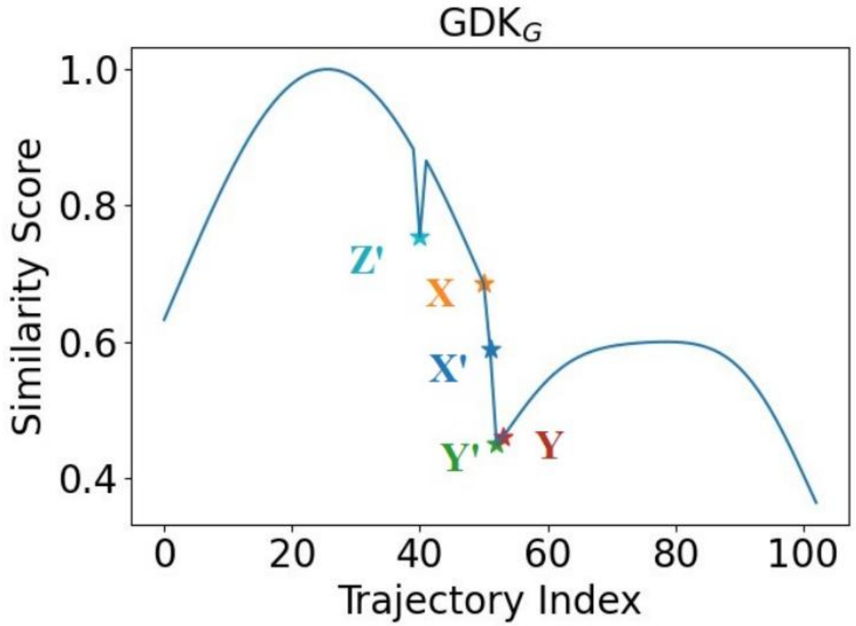


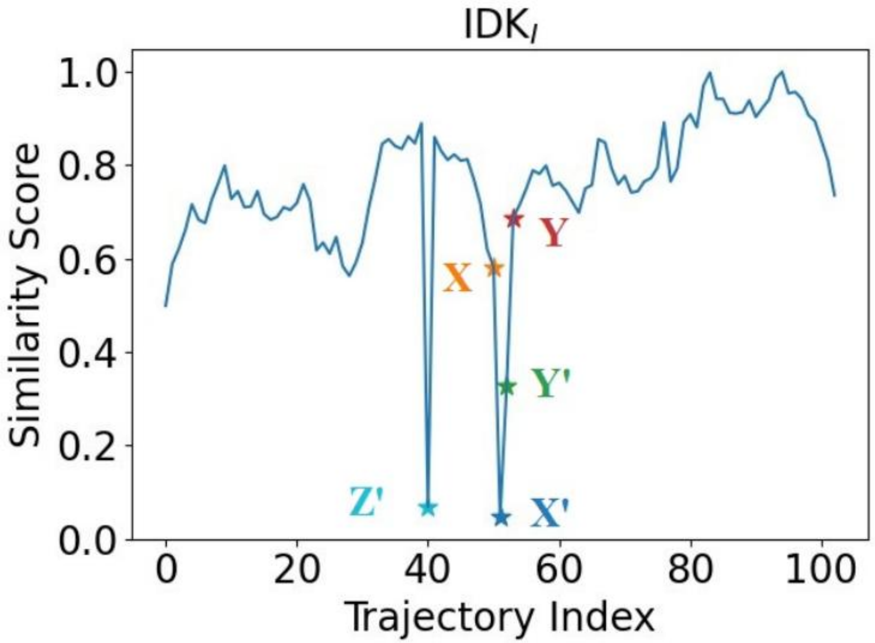
		(X,X')	(X,Y)	$d(X,X') < d(X,Y)$
I	d_W	.50	.43	$\times$
	d_H	.50	.50	$\times$
	d_F	.50	.50	$\times$
II	$\mathcal{K}_I$	.48	.23	✓
	$\mathcal{K}_G$	.45	.27	✓

Data−dependent Property of $\mathcal{K}_I$

- Two distributions are more similar to each other when measured by $\mathcal{K}_I$ derived from a sparse region than that from a dense region.

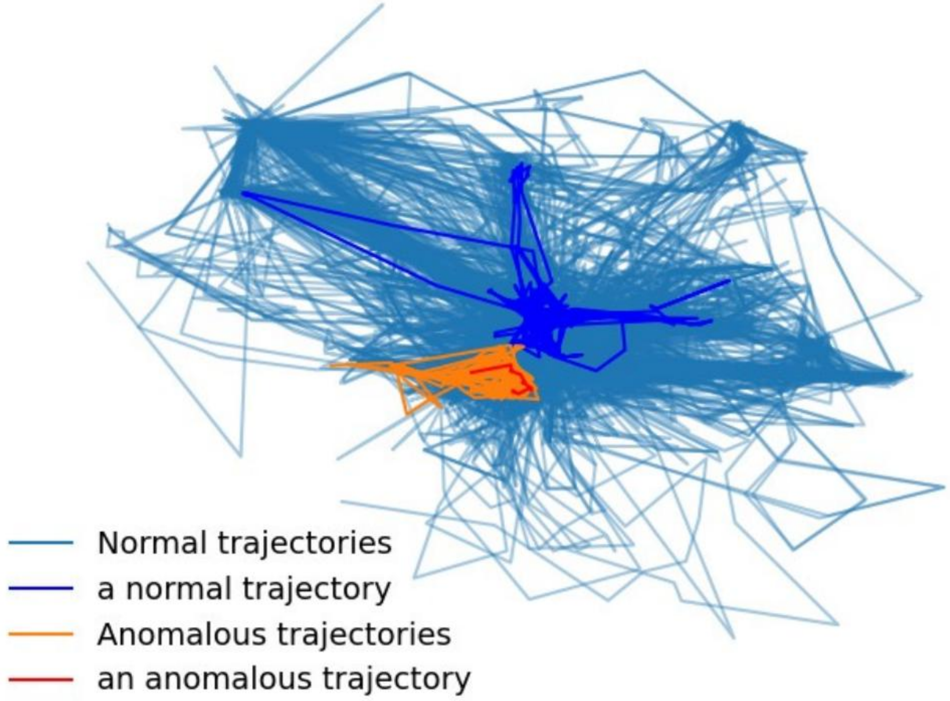






Data−dependent Property of $\mathcal{K}_I$

- *Sheepdogs* Dataset
  - Dense region： sheep (orange)
  - Sparse region： sheepdogs (blue)



ED-based			$\mathcal{K}$AT					
LOF_W	LOF_H	LOF_F	LOF_I	LOF_G	SVM_I	SVM_G	GDK_G	IDK_I
.81	.72	.56	**.99**	.93	.95	.71	.70	.98

Measures for Time Series

Domain/type	Example measures
Time	ℓ_p-norm, DTW, Jaccard, cross-correlation
Frequency	Wasserstein–Fourier distance

- Current measures are in either time or frequency domain.
- Issue : Sensitive to misalignment between subsequences
- Remedies: Dynamic Time Warping (DTW), Shape-based distance (SBD), etc.
- Root cause: the use of a point-to-point distance (high computation cost)

Distance—based Time Series Anomaly Detectors

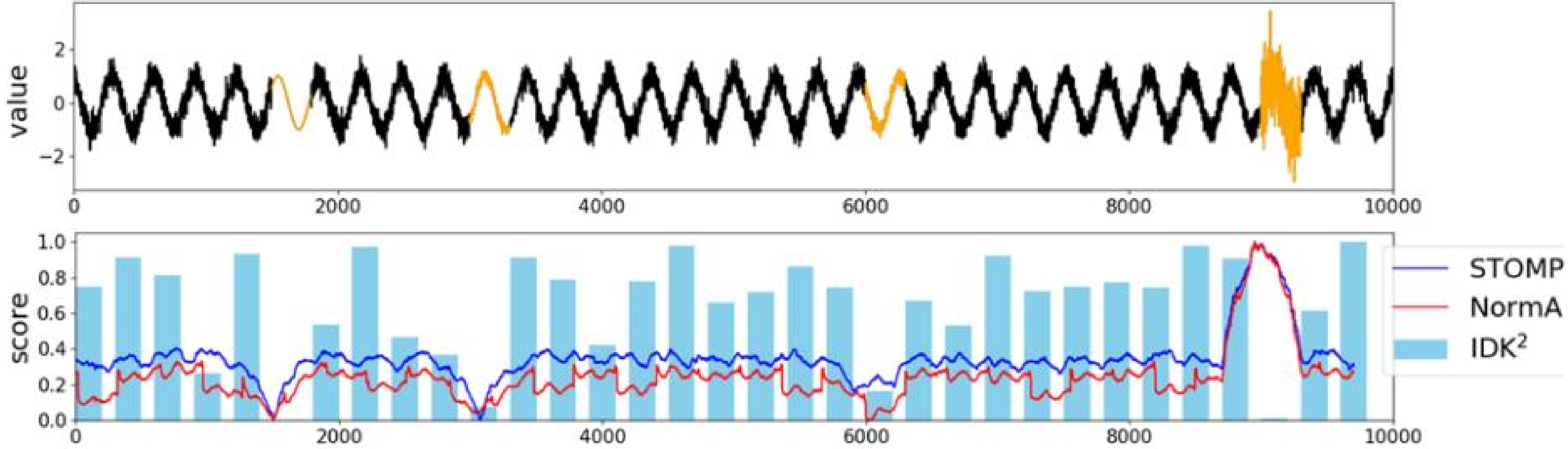
- STOMP computes the z-normalized Euclidean distances between all sequences that can be extracted from a time series and their nearest neighbors in the time series based on a point-to-point distance
- NormA produces a normal model (that consists of mean vectors of clusters of sequences) that can be differentiated from anomalies on a dataset, computing the distance between two sequences based on Euclidean distance.
- Summarize their key approach used to detect anomalies

STOMP: Zhu et. al. Matrix profile II: Exploiting a novel algorithm and GPUs to break the one hundred million barrier for time series motifs and joins. IEEE ICDM, (2016)

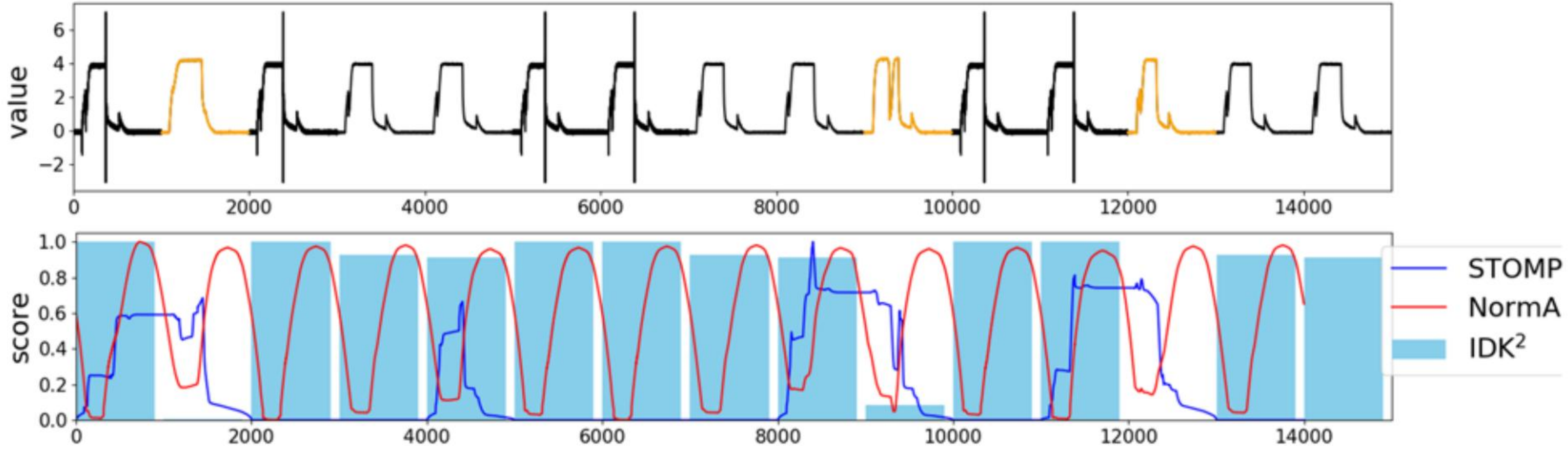
NormA: Boniol et. al. Unsupervised and scalable subsequence anomaly detection in large data series. VLDB J. (2021)

Explain the detection results of (i) STOMP, (ii) NormA, (iii) IDK^2

Noisy_sine



TEK

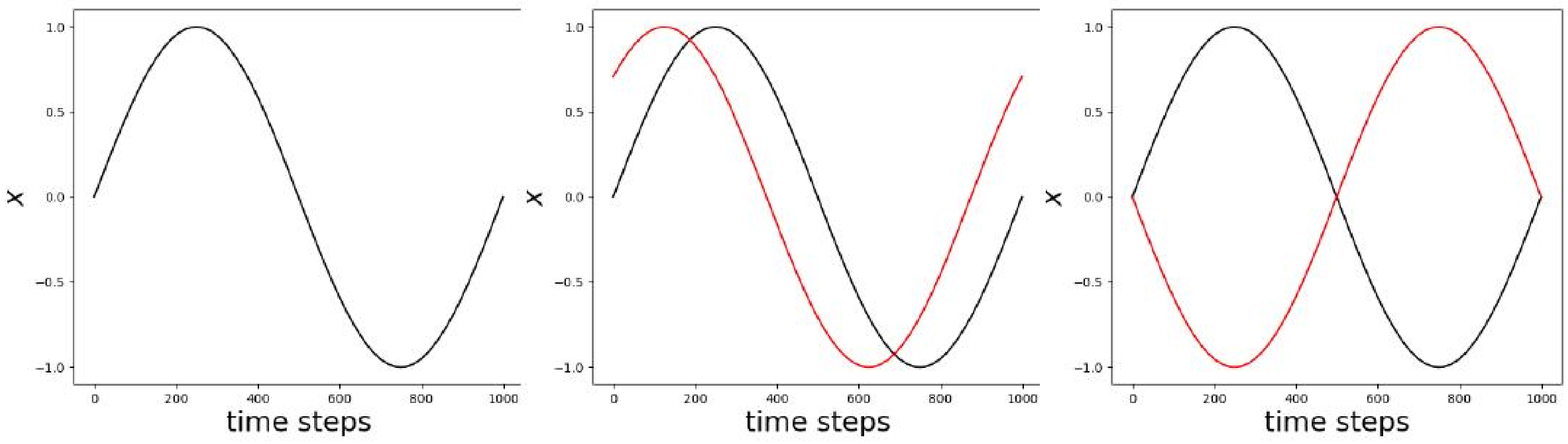


Discussion

- Explain the key differences between IDK^2 and STOMP/NormA

Intuitive Examples

~~Sliding windows are required to find the matched alignment~~



(a) $X_h, h = 0$ **(b)** $X_h, h = 125$ **(c)** $X_h, h = 500$

Distributional versus non-Distributional Measures

Distributional measures	Non-distributional measures
Robust to noise	Sensitive to noise
No alignment issues	Have alignment issues and remedies (DTW, SBD) have additional computational cost

- Existing distributional measures require a computationally expensive process. Both WD and KME have a time cost of at least m^2, where m is the length of each subsequence.
- A recent breakthrough in KME called Isolation Distributional Kernel (IDK) has a time cost of m.

What have you learned?

1. What is the commonly used approach to anomaly detection (e.g., OCSVM, Nearest Neighbor AD, KDE, STOMP, NormA etc)?
2. How does IDK or IDK^2 differ from the above approach?
3. How does IDK^2 differ from IDK or Isolation Forest?
4. How does IDK differ from Isolation Forest?
5. How does IDK differ from iNNE

   (when both employ hyperspheres)?
6. Summarize the key messages.